Computer Networks Lab Ex no. 10

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Comparison of routing protocol (AODV,DSDV,DSR OLSR) using ns3 and plotting the results

1. Copy the manet-routing-compare.cc file from examples/routing to ns-3.43/scratch
2. Navigate to the ns-3 root directory
   cd ns-allinone-3.43/ns-3.43
3. Build the project
   ./ns3 build
4. Ensure no conflicting files (like sum.cc) are present in the scratch/ folder; delete them if necessary
5. Run the base simulation
   ./ns3 run scratch/manet-routing-compare
6. Run simulation for AODV protocol
   ./ns3 run "scratch/manet-routing-compare --CSVfileName=naodv.csv --protocol=AODV"
7. Run simulation for DSR protocol
   ./ns3 run "scratch/manet-routing-compare --CSVfileName=ndsr.csv --protocol=DSR"
8. Run simulation for DSDV protocol
   ./ns3 run "scratch/manet-routing-compare --CSVfileName=ndsdv.csv --protocol=DSDV"
9. Run simulation for OLSR protocol
   ./ns3 run "scratch/manet-routing-compare --CSVfileName=nolsr.csv --protocol=OLSR"
10. Convert CSV files to space-separated format
    tr ',' ' ' < naodv.csv > naodv.dat
    tr ',' ' ' < ndsr.csv > ndsr.dat
    tr ',' ' ' < ndsdv.csv > ndsdv.dat
    tr ',' ' ' < nolsr.csv > nolsr.dat

11. Create a gnuplot script file
nano plot.plot
12. Add the following content inside plot.plot

```
set terminal pdf

set output "compare.pdf"

set xlabel "Time(Sec)"

set ylabel "Receive Rate"

plot "naodv.csv" using 1:2 with points title "AODV" , "ndsdv.csv" using 1:2 with points title "DSDV", "ndsr.csv" using 1:2 with points title "DSR","nolsr.csv" using 1:2 with points title "OLSR"

set xlabel "Time(Sec)"

set ylabel "Packets Received"

plot "naodv.csv" using 1:3 with line title "AODV" , "ndsdv.csv" using 1:3 with line title "DSDV", "ndsr.csv" using 1:3 with line title "DSR","nolsr.csv" using 1:3 with line title "OLSR"
```

13. Save and exit the editor
14. Run gnuplot from terminal
gnuplot plot.plot
15. Open the generated output file
compare.pdf

naodv.csv
~/ns3/ns-allinone-3.43/ns-3.43

naodv.csv | dsdv.csv | olsr.csv | dsr.csv | protocol.plot

```
0 0 0 10 AODV 7.5
1 0 0 10 AODV 7.5
2 0 0 10 AODV 7.5
3 0 0 10 AODV 7.5
4 0 0 10 AODV 7.5
5 0 0 10 AODV 7.5
6 0 0 10 AODV 7.5
7 0 0 10 AODV 7.5
8 0 0 10 AODV 7.5
9 0 0 10 AODV 7.5
10 0 0 10 AODV 7.5
11 0 0 10 AODV 7.5
12 0 0 10 AODV 7.5
13 0 0 10 AODV 7.5
14 0 0 10 AODV 7.5
15 0 0 10 AODV 7.5
16 0 0 10 AODV 7.5
17 0 0 10 AODV 7.5
18 0 0 10 AODV 7.5
19 0 0 10 AODV 7.5
20 0 0 10 AODV 7.5
21 0 0 10 AODV 7.5
22 0 0 10 AODV 7.5
23 0 0 10 AODV 7.5
24 0 0 10 AODV 7.5
25 0 0 10 AODV 7.5
26 0 0 10 AODV 7.5
27 0 0 10 AODV 7.5
28 0 0 10 AODV 7.5
29 0 0 10 AODV 7.5
30 0 0 10 AODV 7.5
31 0 0 10 AODV 7.5
32 0 0 10 AODV 7.5
33 0 0 10 AODV 7.5
34 0 0 10 AODV 7.5
35 0 0 10 AODV 7.5
36 0 0 10 AODV 7.5
37 0 0 10 AODV 7.5
```

CSV | Tab Width: 8 | Ln 29, Col 19 | INS

Open Save

dsdv.csv
~/ns3/ns-allinone-3.43/ns-3.43

naodv.csv | dsdv.csv | olsr.csv | dsr.csv | protocol.plot

```
0 0 0 10 DSDV 7.5
1 0 0 10 DSDV 7.5
2 0 0 10 DSDV 7.5
3 0 0 10 DSDV 7.5
4 0 0 10 DSDV 7.5
5 0 0 10 DSDV 7.5
6 0 0 10 DSDV 7.5
7 0 0 10 DSDV 7.5
8 0 0 10 DSDV 7.5
9 0 0 10 DSDV 7.5
10 0 0 10 DSDV 7.5
11 0 0 10 DSDV 7.5
12 0 0 10 DSDV 7.5
13 0 0 10 DSDV 7.5
14 0 0 10 DSDV 7.5
15 0 0 10 DSDV 7.5
16 0 0 10 DSDV 7.5
17 0 0 10 DSDV 7.5
18 0 0 10 DSDV 7.5
19 0 0 10 DSDV 7.5
20 0 0 10 DSDV 7.5
21 0 0 10 DSDV 7.5
22 0 0 10 DSDV 7.5
23 0 0 10 DSDV 7.5
24 0 0 10 DSDV 7.5
25 0 0 10 DSDV 7.5
26 0 0 10 DSDV 7.5
27 0 0 10 DSDV 7.5
28 0 0 10 DSDV 7.5
29 0 0 10 DSDV 7.5
30 0 0 10 DSDV 7.5
31 0 0 10 DSDV 7.5
32 0 0 10 DSDV 7.5
33 0 0 10 DSDV 7.5
34 0 0 10 DSDV 7.5
35 0 0 10 DSDV 7.5
36 0 0 10 DSDV 7.5
37 0 0 10 DSDV 7.5
```

CSV Tab Width: 8 Ln 1, Col 1 INS

Open Save

olsr.csv
~/ns3/ns-allinone-3.43/ns-3.43

naodv.csv dsdv.csv olsr.csv dsr.csv protocol.plot

```
0 0 0 10 OLSR 7.5
1 0 0 10 OLSR 7.5
2 0 0 10 OLSR 7.5
3 0 0 10 OLSR 7.5
4 0 0 10 OLSR 7.5
5 0 0 10 OLSR 7.5
6 0 0 10 OLSR 7.5
7 0 0 10 OLSR 7.5
8 0 0 10 OLSR 7.5
9 0 0 10 OLSR 7.5
10 0 0 10 OLSR 7.5
11 0 0 10 OLSR 7.5
12 0 0 10 OLSR 7.5
13 0 0 10 OLSR 7.5
14 0 0 10 OLSR 7.5
15 0 0 10 OLSR 7.5
16 0 0 10 OLSR 7.5
17 0 0 10 OLSR 7.5
18 0 0 10 OLSR 7.5
19 0 0 10 OLSR 7.5
20 0 0 10 OLSR 7.5
21 0 0 10 OLSR 7.5
22 0 0 10 OLSR 7.5
23 0 0 10 OLSR 7.5
24 0 0 10 OLSR 7.5
25 0 0 10 OLSR 7.5
26 0 0 10 OLSR 7.5
27 0 0 10 OLSR 7.5
28 0 0 10 OLSR 7.5
29 0 0 10 OLSR 7.5
30 0 0 10 OLSR 7.5
31 0 0 10 OLSR 7.5
32 0 0 10 OLSR 7.5
33 0 0 10 OLSR 7.5
34 0 0 10 OLSR 7.5
35 0 0 10 OLSR 7.5
36 0 0 10 OLSR 7.5
37 0 0 10 OLSR 7.5
```

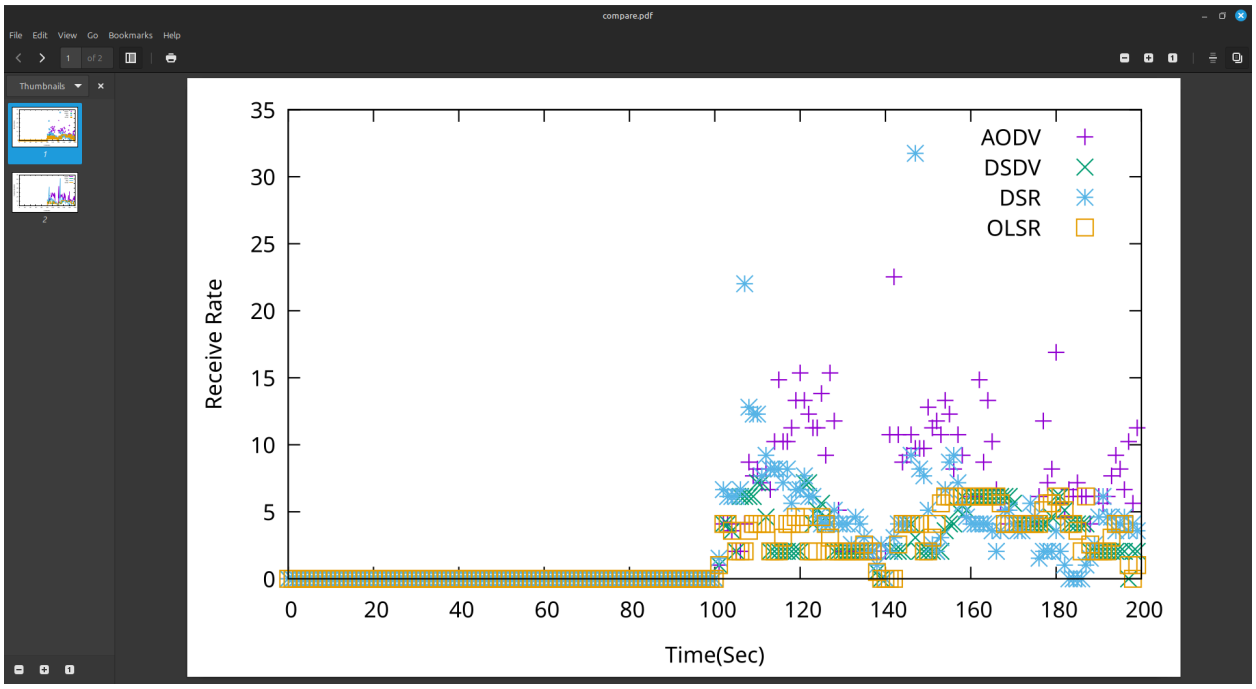
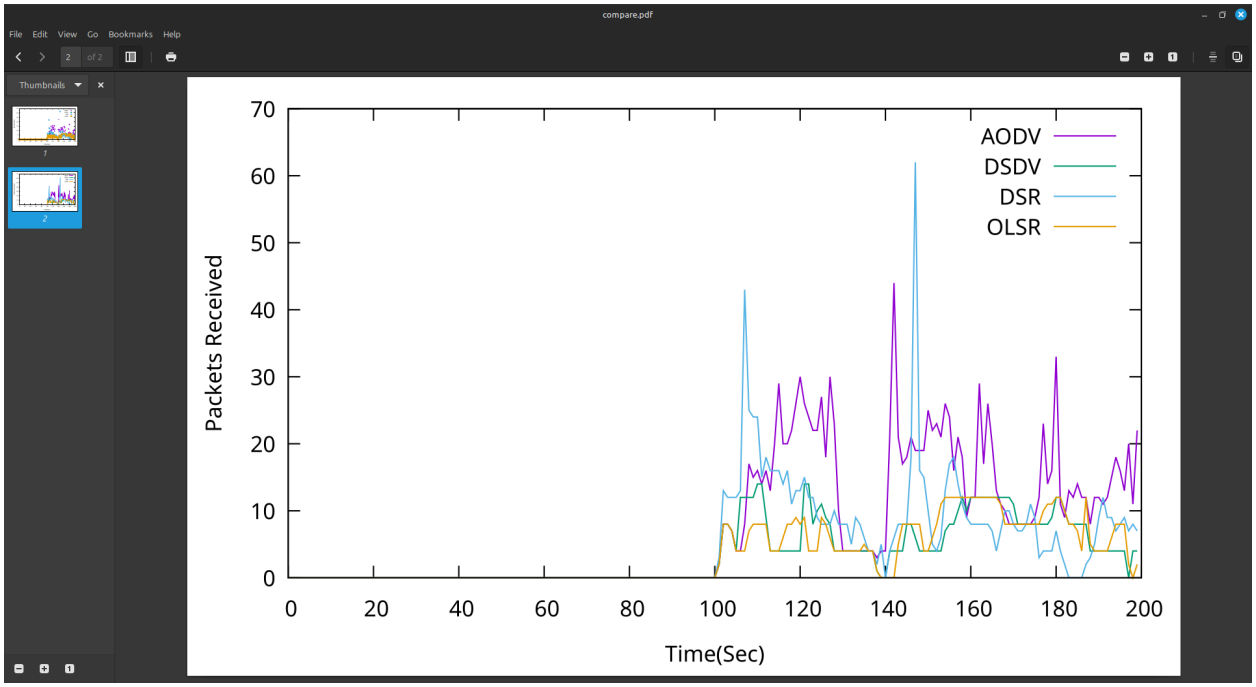
CSV Tab Width: 8 Ln 181, Col 25 INS

Open Save

dsr.csv
~/ns3/ns-allinone-3.43/ns-3.43

naodv.csv dsdv.csv olsr.csv dsr.csv protocol.plot

```csv
0 0 0 10 DSR 7.5
1 0 0 10 DSR 7.5
2 0 0 10 DSR 7.5
3 0 0 10 DSR 7.5
4 0 0 10 DSR 7.5
5 0 0 10 DSR 7.5
6 0 0 10 DSR 7.5
7 0 0 10 DSR 7.5
8 0 0 10 DSR 7.5
9 0 0 10 DSR 7.5
10 0 0 10 DSR 7.5
11 0 0 10 DSR 7.5
12 0 0 10 DSR 7.5
13 0 0 10 DSR 7.5
14 0 0 10 DSR 7.5
15 0 0 10 DSR 7.5
16 0 0 10 DSR 7.5
17 0 0 10 DSR 7.5
18 0 0 10 DSR 7.5
19 0 0 10 DSR 7.5
20 0 0 10 DSR 7.5
21 0 0 10 DSR 7.5
22 0 0 10 DSR 7.5
23 0 0 10 DSR 7.5
24 0 0 10 DSR 7.5
25 0 0 10 DSR 7.5
26 0 0 10 DSR 7.5
27 0 0 10 DSR 7.5
28 0 0 10 DSR 7.5
29 0 0 10 DSR 7.5
30 0 0 10 DSR 7.5
31 0 0 10 DSR 7.5
32 0 0 10 DSR 7.5
33 0 0 10 DSR 7.5
34 0 0 10 DSR 7.5
35 0 0 10 DSR 7.5
36 0 0 10 DSR 7.5
37 0 0 10 DSR 7.5
```

CSV Tab Width: 8 Ln 25, Col 18 INS

Comparison Graphs